Empirical Estimation on Orbital Spaces: Glivenko-Cantelli and Donsker Theorems under Group Actions with Applications to Equivariant Density Estimation

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Abstract

We develop a comprehensive theory of empirical estimation when the sample space is partitioned into orbits under a discrete group action. Given a topological space E with action of $G = \langle \varphi \rangle$, we construct orbital empirical measures on the quotient space E/G and establish their convergence properties. Our main results include: (1) an Orbital Glivenko-Cantelli theorem with rates adapted to the twin metric d_G; (2) an Orbital Donsker theorem for functional convergence of the empirical process; (3) equivariant kernel density estimation with optimal bandwidth selection respecting the orbital structure; (4) maximum likelihood theory on quotient spaces with orbital Fisher information. We show that exploiting the group symmetry reduces the effective dimension on the quotient and can lower the variance of orbit-averaged estimators by up to a factor equal to the orbit size, and we establish concentration inequalities for orbital statistics. (For a G-invariant model the orbit is already sufficient, so symmetry does not inflate the effective sample size; the gains are in dimension and in variance reduction of non-invariant functionals.) Applications include directional statistics, compositional data analysis, and estimation problems with physical symmetries.

Keywords: Orbital empirical measures; Glivenko-Cantelli theorem; Donsker theorem; Equivariant estimation; Quotient spaces; Group actions; Kernel density estimation; Fisher information; Directional statistics
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Contents

1 Introduction

1.1 Motivation: Statistics with Symmetries

Many statistical problems possess inherent symmetries (Eaton, 1989; Wijsman, 1990). When measuring angles, the data live on a circle with rotational symmetry. When analyzing proportions, the data live on a simplex with permutation structure (Aitchison, 1986). In physics, measurements may be invariant under gauge transformations or coordinate changes.

Classical statistical theory typically ignores these symmetries, treating observations as points in Euclidean space (van der Vaart, 1998). This approach has two drawbacks:

(i) It fails to exploit the symmetry structure for improved estimation

(ii) It may produce estimators that are not equivariant under the natural transformations

The theory of Twin Spaces provides a framework for statistics adapted to non-Euclidean structures (Patrangenaru and Ellingson, 2016; Pennec, 2006). In this article, we extend this framework to spaces with group actions, where the natural objects of study are orbits rather than individual points.

1.2 Setting and Notation

Let E be a topological space (typically a metric space or manifold) and let $G = \langle \varphi \rangle \cong \mathbb{Z}$ be a discrete group acting continuously on E:

$$(n, x) \mapsto \varphi^n(x), \quad n \in \mathbb{Z}, \quad x \in E. \tag{1}$$

The orbit of $x \in E$ is:

$$O(x) := \{\varphi^n(x) : n \in \mathbb{Z}\}. \tag{2}$$

The orbit space (or quotient space) is:

$$E/G := \{O(x) : x \in E\} \tag{3}$$

equipped with the quotient topology induced by the projection $\pi : E \to E/G$.

1.3 Key Questions

Given i.i.d. observations $X_1, \ldots, X_n$ from a distribution μ on E:

1. How do we estimate the induced measure μ_G on E/G?
2. What is the convergence rate in the orbital metric d_G?
3. How does the Twin Stone-Weierstrass theorem inform function approximation?
4. Can we improve estimation by averaging over orbits?
5. What is the orbital analogue of Fisher information?

1.4 Main Results Overview

1. **Orbital Glivenko-Cantelli** (Theorem 4.3): The orbital empirical measure $\hat{\mu}_n^G$ converges almost surely to μ_G in the twin-sup norm.
2. **Orbital Donsker** (Theorem 5.3): The orbital empirical process converges weakly to a Gaussian process indexed by $C_{\text{twin}}(E)$.
3. **Equivariant Kernel Estimation** (Theorem 6.6): Kernel density estimators using equivariant kernels achieve optimal rates on E/G.
4. **Orbital MLE** (Theorem 7.5): The MLE on quotient spaces is asymptotically normal with variance given by inverse orbital Fisher information.
5. **Orbit-Averaging Improvement** (Theorem 6.8): Averaging over finite orbits reduces variance by up to a factor of $|G|$ (achieved when orbit points are separated by more than the bandwidth).

1.5 Related Work

The foundations of invariant statistical procedures were established by Hunt and Stein (1966) and Lehmann and Casella (1998), showing when estimators should be equivariant under group actions. Stein (1956) proved that the usual estimator of a multivariate normal mean is inadmissible when the dimension exceeds 2, a result related to the geometry of spheres.

Directional statistics was pioneered by Fisher (1953) and systematized by Mardia and Jupp (2000). The theory of empirical processes was developed by Glivenko (1933), Cantelli (1933), and Donsker (1952), with modern treatments by Dudley (1999) and van der Vaart and Wellner (1996). Information geometry was founded by Amari (2016), providing geometric frameworks for statistical manifolds.

Our contribution is to synthesize these traditions within the Twin Space framework, establishing unified convergence theory for orbital estimation.

2 Twin Topology and Orbital Geometry

We recall the essential definitions from the approximation theory on Twin Spaces.

2.1 Twin Metric

Definition 2.1 (Twin Metric)**.** *Assume (E, d) is a metric space and φ is an isometry. The twin metric (or orbital metric) on E/G is the infimum of the distance over relative group shifts:*

$$d_G(O(x), O(y)) := \inf_{n \in \mathbb{Z}} d(x, \varphi^n y) = \inf_{m,n \in \mathbb{Z}} d(\varphi^m x, \varphi^n y). \tag{4}$$

The two infima agree because φ is an isometry, so $d(\varphi^m x, \varphi^n y) = d(x, \varphi^{n-m} y)$. When orbits are finite (e.g., G is a finite group), the infimum is a minimum over finitely many terms.

Remark 2.2 (Why the infimum, not the supremum)**.** *The infimum over relative shifts is forced by well-definedness on the quotient: replacing the representative y by $\varphi^j y$ leaves $\inf_n d(x, \varphi^n y)$ unchanged, whereas $\sup_n d(\varphi^n x, \varphi^n y)$ would equal $d(x, y)$ for an isometry φ and would therefore depend on the chosen representatives (and fail $d_G(O(x), O(x)) = 0$). Equation (4) is the standard quotient (orbit) metric and is the one used in the computational sections (cf. (57), (60)).*

Proposition 2.3 (Metric Properties)**.** *If the orbits of G are closed in E, then d_G is a well-defined metric on E/G satisfying:*

(i) $d_G(O(x), O(y)) = 0$ *iff* $O(x) = O(y)$

(ii) $d_G(O(x), O(y)) = d_G(O(y), O(x))$

(iii) $d_G(O(x), O(z)) \leq d_G(O(x), O(y)) + d_G(O(y), O(z))$

Proof. Well-definedness on orbits is Remark 2.2.

(i) If $d_G(O(x), O(y)) = 0$ then $\inf_n d(x, \varphi^n y) = 0$; since the orbit $O(y)$ is closed, the infimum is attained (or approached within $O(y)$), giving $x = \varphi^n y$ for some n, i.e. $O(x) = O(y)$. The converse is immediate.

(ii) Using that φ is an isometry, $d_G(O(x), O(y)) = \inf_n d(x, \varphi^n y) = \inf_n d(\varphi^{-n} x, y) = \inf_m d(y, \varphi^m x) = d_G(O(y), O(x))$.

(iii) For any $n, k \in \mathbb{Z}$, the isometry property gives $d(x, \varphi^n z) \leq d(x, \varphi^k y) + d(\varphi^k y, \varphi^n z) = d(x, \varphi^k y) + d(y, \varphi^{n-k} z)$. Taking the infimum over n and then over k yields the triangle inequality. □

2.2 Twin Continuity

Definition 2.4 (Twin-Continuous Function)**.** *A function $f : E \to \mathbb{R}$ is twin-continuous if:*

1. *f is continuous on E;*
2. *For every compact $K \subset E$, the family $\{f \circ \varphi^n\}_{n \in \mathbb{Z}}$ is equicontinuous on K.*

The space of twin-continuous functions is denoted $C_{\text{twin}}(E)$.

Definition 2.5 (Twin-Sup Norm)**.** *The twin-sup norm on $C_{\text{twin}}(E)$ is:*

$$\|f\|_{\infty,\text{twin}} := \sup_{x \in E} \sup_{n \in \mathbb{Z}} |f(\varphi^n x)|. \tag{5}$$

Lemma 2.6 (Banach Space Structure)**.** *$(C_{\text{twin}}(E), \|\cdot\|_{\infty,\text{twin}})$ is a Banach space.*

2.3 Equivariant Functions

Definition 2.7 (G-Invariant Function). *A function $f : E \to \mathbb{R}$ is G-invariant if:*

$$f(\varphi^n x) = f(x) \quad \forall x \in E, \forall n \in \mathbb{Z}. \tag{6}$$

Equivalently, f factors through the quotient: $f = \tilde{f} \circ \pi$ for some $\tilde{f} : E/G \to \mathbb{R}$.

Definition 2.8 (G-Equivariant Function). *A function $f : E \to F$ (where G also acts on F) is G-equivariant if:*

$$f(\varphi^n x) = \varphi^n f(x) \quad \forall x \in E, \forall n \in \mathbb{Z}. \tag{7}$$

The space of G-invariant twin-continuous functions is denoted $C^G_{\mathrm{twin}}(E)$.

2.4 The Twin Stone-Weierstrass Theorem

The following fundamental result guarantees density of polynomial algebras (Stone, 1948; Nachbin, 1965).

Theorem 2.9 (Twin Stone-Weierstrass). *Let (E, G) be a compact Twin Space and let $\mathcal{A}$ be the algebra generated by twin-equivariant polynomials. Then $\mathcal{A}$ is dense in $C_{\mathrm{twin}}(E)$ with respect to $\|\cdot\|_{\infty,\mathrm{twin}}$.*

This theorem is crucial for approximation-theoretic arguments in estimation.

3 Orbital Empirical Measures

Let $X_1, \dots, X_n$ be i.i.d. observations from a probability measure μ on E.

Definition 3.1 (Empirical Measure). *The empirical measure on E is:*

$$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}. \tag{8}$$

Definition 3.2 (Orbital Empirical Measure). *The orbital empirical measure on E/G is:*

$$\hat{\mu}^G_n := \frac{1}{n} \sum_{i=1}^{n} \delta_{O(X_i)} = \pi_* \hat{\mu}_n. \tag{9}$$

This is the pushforward of $\hat{\mu}_n$ under the quotient map $\pi : E \to E/G$.

Definition 3.3 (Orbit-Averaged Empirical Measure). *When G is finite with $|G| = m$, the orbit-averaged empirical measure is:*

$$\hat{\mu}^{\mathrm{avg}}_n := \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m} \sum_{g \in G} \delta_{g \cdot X_i} = \frac{1}{nm} \sum_{i=1}^{n} \sum_{k=0}^{m-1} \delta_{\varphi^k(X_i)}. \tag{10}$$

Proposition 3.4 (Measure Relationships). *Let $\mu_G := \pi_* \mu$ be the pushforward of μ to E/G. Then:*

(i) $\mathbb{E}[\hat{\mu}^G_n] = \mu_G$

(ii) For G-invariant f: $\int f \, d\hat{\mu}^G_n = \int f \, d\hat{\mu}_n$

(iii) If μ is G-invariant: $\mathbb{E}[\hat{\mu}_n^{\mathrm{avg}}] = \mu$

Proof. (i) For any measurable $A \subset E/G$:

$$\mathbb{E}[\hat{\mu}_n^G(A)] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \mathbf{1}_{O(X_i)\in A}\right] = P(O(X_1) \in A) = \mu_G(A). \tag{11}$$

(ii) If $f = \tilde{f} \circ \pi$ is G-invariant:

$$\int f \, d\hat{\mu}_n^G = \frac{1}{n}\sum_{i=1}^n \tilde{f}(O(X_i)) = \frac{1}{n}\sum_{i=1}^n f(X_i) = \int f \, d\hat{\mu}_n. \tag{12}$$

(iii) Follows from G-invariance of μ. □

4 Orbital Glivenko-Cantelli Theory

4.1 Classical Glivenko-Cantelli

Recall the classical result (Glivenko, 1933; Cantelli, 1933): for $X_1, \ldots, X_n$ i.i.d. from F on $\mathbb{R}$,

$$\sup_t |F_n(t) - F(t)| \xrightarrow{a.s.} 0 \tag{13}$$

where $F_n(t) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}_{X_i \le t}$.

4.2 Glivenko-Cantelli Classes

Definition 4.1 (Glivenko-Cantelli Class)**.** *A class of functions $\mathcal{F}$ on E is a Glivenko-Cantelli class for μ if:*

$$\sup_{f\in\mathcal{F}} \left| \int f \, d\hat{\mu}_n - \int f \, d\mu \right| \xrightarrow{a.s.} 0. \tag{14}$$

Definition 4.2 (Orbital Glivenko-Cantelli Class)**.** *A class $\mathcal{F} \subset C^G_{\mathrm{twin}}(E)$ of G-invariant functions is an orbital Glivenko-Cantelli class if:*

$$\sup_{f\in\mathcal{F}} \left| \int f \, d\hat{\mu}_n^G - \int f \, d\mu_G \right| \xrightarrow{a.s.} 0. \tag{15}$$

4.3 Main Glivenko-Cantelli Result

Theorem 4.3 (Orbital Glivenko-Cantelli)**.** *Let (E, G) be a compact Twin Space with G acting by isometries. Let μ be a probability measure on E. Then the class $\mathcal{F} = \{f \in C^G_{\mathrm{twin}}(E) : \|f\|_{\infty,\mathrm{twin}} \le 1\}$ of bounded G-invariant twin-continuous functions is an orbital Glivenko-Cantelli class:*

$$\sup_{\|f\|_{\infty,\mathrm{twin}}\le 1, f\in C^G_{\mathrm{twin}}(E)} \left| \int f \, d\hat{\mu}_n^G - \int f \, d\mu_G \right| \xrightarrow{a.s.} 0. \tag{16}$$

Proof. The proof proceeds in three steps.

Step 1: Reduction to quotient space. Since f is G-invariant, $f = \tilde{f} \circ \pi$ for some continuous $\tilde{f} : E/G \to \mathbb{R}$. The integral becomes:

$$\int f \, d\hat{\mu}_n^G = \int \tilde{f} \, d(\pi_* \hat{\mu}_n) = \frac{1}{n}\sum_{i=1}^n \tilde{f}(O(X_i)). \tag{17}$$

Step 2: Compactness of E/G. Since E is compact and π is continuous and surjective, E/G is compact.

Step 3: Apply classical theory. The observations $O(X_1), \dots, O(X_n)$ are i.i.d. from μ_G on the compact metric space $(E/G, d_G)$. By the classical Glivenko-Cantelli theorem for compact metric spaces (Dudley, 1999), the class of continuous functions bounded by 1 is a GC class.

Since $\tilde{f}$ is continuous on E/G whenever $f \in C^G_{\text{twin}}(E)$, the result follows. □

4.4 Convergence Rate

Theorem 4.4 (Orbital Glivenko-Cantelli Rate). *Under the conditions of Theorem 4.3, for any $\epsilon > 0$:*

$$P\left(\sup_{f\in\mathcal{F}}\left|\int f\, d\hat{\mu}^G_n - \int f\, d\mu_G\right| > \epsilon\right) \le 2N(\epsilon/2, \mathcal{F}, \|\cdot\|_{\infty,\text{twin}}) \cdot e^{-n\epsilon^2/8} \tag{18}$$

where $N(\epsilon, \mathcal{F}, \|\cdot\|)$ is the covering number of $\mathcal{F}$.

Proof. Apply the standard symmetrization and chaining arguments (van der Vaart and Wellner, 1996) to the class $\tilde{\mathcal{F}} = \{\tilde{f} : f \in \mathcal{F}\}$ on E/G, using the metric entropy of the function class. □

Corollary 4.5 (Rate for Lipschitz Classes). *If $\mathcal{F}$ consists of G-invariant functions that are L-Lipschitz in the twin metric, and $(E/G, d_G)$ has covering dimension d, then:*

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\int f\, d\hat{\mu}^G_n - \int f\, d\mu_G\right|\right] = O\left(n^{-1/(2\vee d)}\right). \tag{19}$$

5 Orbital Donsker Theory

5.1 The Orbital Empirical Process

Definition 5.1 (Orbital Empirical Process). *The orbital empirical process indexed by $\mathcal{F} \subset C^G_{\text{twin}}(E)$ is:*

$$\mathbb{G}^G_n(f) := \sqrt{n}\left(\int f\, d\hat{\mu}^G_n - \int f\, d\mu_G\right) = \frac{1}{\sqrt{n}}\sum_{i=1}^n (f(X_i) - \mathbb{E}[f(X_1)]). \tag{20}$$

Note that for G-invariant f, this equals the classical empirical process evaluated at f.

5.2 Donsker Classes

Definition 5.2 (Orbital Donsker Class). *A class $\mathcal{F} \subset C^G_{\text{twin}}(E)$ is an orbital Donsker class if $\mathbb{G}^G_n$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process $\mathbb{G}^G$.*

Theorem 5.3 (Orbital Donsker). *Let (E, G) be a compact Twin Space. Let $\mathcal{F} \subset C^G_{\text{twin}}(E)$ be a class of G-invariant functions with envelope F satisfying $\|F\|_{L^2(\mu)} < \infty$. If the bracketing entropy satisfies:*

$$\int_0^\infty \sqrt{\log N_{[\,]}(\epsilon, \mathcal{F}, L^2(\mu_G))}\, d\epsilon < \infty, \tag{21}$$

then $\mathcal{F}$ is an orbital Donsker class, and:

$$\mathbb{G}^G_n \Rightarrow \mathbb{G}^G \quad \text{in } \ell^\infty(\mathcal{F}), \tag{22}$$

where $\mathbb{G}^G$ is a centered Gaussian process with covariance:

$$\mathbb{E}[\mathbb{G}^G(f)\mathbb{G}^G(g)] = \int fg\, d\mu_G - \int f\, d\mu_G \int g\, d\mu_G. \tag{23}$$

Proof. The proof follows from the classical Donsker theorem (Donsker, 1952; van der Vaart and Wellner, 1996) applied to the i.i.d. sequence $O(X_1), \dots, O(X_n)$ on $(E/G, \mu_G)$, using the correspondence between G-invariant functions on E and functions on E/G. □

5.3 Application: Kolmogorov-Smirnov Test on Orbits

Corollary 5.4 (Orbital KS Statistic). *Define the orbital Kolmogorov-Smirnov statistic:*

$$D_n^G := \sup_{f \in \mathcal{F}_{\text{ind}}} |\mathbb{G}_n^G(f)| \tag{24}$$

where $\mathcal{F}_{\text{ind}}$ is the class of indicators of G-invariant sets. Under the null hypothesis $\mu = \mu_0$:

$$D_n^G \Rightarrow \sup_{t \in [0,1]} |B(t)| \tag{25}$$

where B is a Brownian bridge (when E/G is one-dimensional).

6 Equivariant Kernel Density Estimation

6.1 Equivariant Kernels

Definition 6.1 (Equivariant Kernel). *A kernel $K : E \times E \to \mathbb{R}$ is G-equivariant if:*

$$K(\varphi^n x, \varphi^n y) = K(x, y) \quad \forall x, y \in E, \forall n \in \mathbb{Z}. \tag{26}$$

Lemma 6.2 (Equivariant Kernel Induces Quotient Kernel). *If K is G-equivariant, then K induces a well-defined kernel $\tilde{K}$ on E/G:*

$$\tilde{K}(O(x), O(y)) := K(x, y). \tag{27}$$

6.2 Orbital Kernel Density Estimator

Definition 6.3 (Orbital KDE). *Given an equivariant kernel K_h with bandwidth $h > 0$, the orbital kernel density estimator is:*

$$\hat{f}_n^G(O(x)) := \frac{1}{n} \sum_{i=1}^n K_h(x, X_i) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{d_G(O(x), O(X_i))}{h}\right) \tag{28}$$

where K is a univariate kernel and d is the dimension of E/G.

Since K_h is equivariant, $\hat{f}_n^G$ is well-defined on E/G.

6.3 Bias and Variance

Proposition 6.4 (Orbital KDE Bias). *Assume the density f_G of μ_G with respect to a reference measure on E/G is twice continuously differentiable. Then:*

$$\mathbb{E}[\hat{f}_n^G(O(x))] = f_G(O(x)) + \frac{h^2}{2}\mu_2(K)\Delta_G f_G(O(x)) + o(h^2) \tag{29}$$

where $\mu_2(K) = \int t^2 K(t)\, dt$ and Δ_G is the Laplacian on E/G.

Proposition 6.5 (Orbital KDE Variance)**.**

$$\mathrm{Var}(\hat{f}_n^G(O(x))) = \frac{f_G(O(x))}{nh^d}\int K^2(t)\,dt + o\left(\frac{1}{nh^d}\right). \tag{30}$$

Theorem 6.6 (Optimal Bandwidth for Orbital KDE)**.** *The mean integrated squared error (MISE) optimal bandwidth is:*

$$h_{\mathrm{opt}}^G = \left(\frac{d\int K^2}{\mu_2(K)^2\int(\Delta_G f_G)^2}\right)^{1/(d+4)} n^{-1/(d+4)} \tag{31}$$

achieving MISE rate $O(n^{-4/(d+4)})$.

6.4 Orbit-Averaged Kernel Estimator

When G is finite, we can average over the orbit to reduce variance.

Definition 6.7 (Orbit-Averaged KDE)**.** *For finite G with $|G| = m$:*

$$\hat{f}_n^{\mathrm{avg}}(x) := \frac{1}{nm}\sum_{i=1}^{n}\sum_{k=0}^{m-1} K_h(x, \varphi^k(X_i)). \tag{32}$$

Theorem 6.8 (Variance Reduction by Orbit Averaging)**.** *If μ is G-invariant, then the orbit-averaged estimator never increases the variance,*

$$\mathrm{Var}(\hat{f}_n^{\mathrm{avg}}(x)) \le \mathrm{Var}(\hat{f}_n(x)), \tag{33}$$

where $\hat{f}_n$ is the standard KDE. Moreover, when the orbit points $\{\varphi^k x\}_{k=0}^{m-1}$ are separated by more than the bandwidth h (so the shifted estimators are asymptotically uncorrelated), the maximal m-fold reduction is achieved,

$$\mathrm{Var}(\hat{f}_n^{\mathrm{avg}}(x)) = \frac{1}{m}\mathrm{Var}(\hat{f}_n(x))\,(1 + o(1)). \tag{34}$$

In general the reduction factor lies between 1 and m.

Proof. Write $\hat{f}_n^{\mathrm{avg}}(x) = \frac{1}{m}\sum_{k=0}^{m-1}\hat{f}_n(\varphi^{-k}(x))$. By G-invariance of μ, each $Y_k := \hat{f}_n(\varphi^{-k}(x))$ has the same variance $V := \mathrm{Var}(\hat{f}_n(x))$, and

$$\mathrm{Var}(\hat{f}_n^{\mathrm{avg}}(x)) = \frac{1}{m^2}\sum_{k,\ell}\mathrm{Cov}(Y_k, Y_\ell) = \frac{1}{m}V + \frac{1}{m^2}\sum_{k\ne\ell}\mathrm{Cov}(Y_k, Y_\ell). \tag{35}$$

By Cauchy–Schwarz $\mathrm{Cov}(Y_k, Y_\ell) \le V$, so $\mathrm{Var}(\hat{f}_n^{\mathrm{avg}}(x)) \le \frac{1}{m^2}\, m^2 V = V$, which is (33). When the orbit points lie more than a bandwidth apart, the kernels $K_h(\,\cdot\,, \varphi^{-k}x)$ and $K_h(\,\cdot\,, \varphi^{-\ell}x)$ have asymptotically disjoint support, so $\mathrm{Cov}(Y_k, Y_\ell) = o(V)$ for $k \ne \ell$ and only the diagonal term $\frac{1}{m}V$ survives, giving (34). Conversely, when orbit points fall within a bandwidth the cross-covariances are positive, so $\mathrm{Var}(\hat{f}_n^{\mathrm{avg}}(x)) \ge \frac{1}{m}V$ and the reduction is less than m-fold. □

7 Maximum Likelihood on Quotient Spaces

7.1 Setup

Consider a parametric family $\{p_\theta : \theta \in \Theta\}$ of densities on E that are G-invariant:

$$p_\theta(\varphi^n x) = p_\theta(x) \quad \forall x, n, \theta. \tag{36}$$

This is equivalent to specifying a parametric family on E/G.

7.2 Orbital Log-Likelihood

Definition 7.1 (Orbital Log-Likelihood)**.** *The orbital log-likelihood based on observations $X_1, \ldots, X_n$ is:*

$$\ell_n^G(\theta) := \sum_{i=1}^n \log p_\theta(X_i) = \sum_{i=1}^n \log \tilde{p}_\theta(O(X_i)) \tag{37}$$

where $\tilde{p}_\theta$ is the induced density on E/G.

7.3 Orbital Fisher Information

Definition 7.2 (Orbital Fisher Information)**.** *The orbital Fisher information is:*

$$I^G(\theta) := \mathbb{E}_\theta \left[\nabla_\theta \log \tilde{p}_\theta(O(X)) \cdot \nabla_\theta \log \tilde{p}_\theta(O(X))^T \right]. \tag{38}$$

Proposition 7.3 (Orbital vs. Original Fisher Information)**.** *If p_θ is G-invariant, then:*

$$I^G(\theta) = I(\theta) \tag{39}$$

where $I(\theta)$ is the classical Fisher information computed from p_θ on E.

Proof. Since p_θ is G-invariant, $\nabla_\theta \log p_\theta(x) = \nabla_\theta \log \tilde{p}_\theta(O(x))$. The expectation over $X \sim p_\theta$ equals the expectation over $O(X) \sim \tilde{p}_\theta$. □

7.4 Asymptotic Theory for Orbital MLE

Theorem 7.4 (Orbital MLE Consistency)**.** *Let $\hat{\theta}_n^G := \arg\max_\theta \ell_n^G(\theta)$. Under standard regularity conditions (identifiability, compactness of Θ, continuity of ℓ_n^G):*

$$\hat{\theta}_n^G \xrightarrow{a.s.} \theta_0 \tag{40}$$

where θ_0 is the true parameter.

Theorem 7.5 (Orbital MLE Asymptotic Normality)**.** *Under regularity conditions:*

$$\sqrt{n}(\hat{\theta}_n^G - \theta_0) \xrightarrow{d} N(0, I^G(\theta_0)^{-1}). \tag{41}$$

Proof. This is the standard MLE asymptotic normality (van der Vaart, 1998) applied to the i.i.d. observations $O(X_1), \ldots, O(X_n)$ from $\tilde{p}_{\theta_0}$ on E/G. □

7.5 Improved Estimation via Orbit Structure

Theorem 7.6 (No spurious gain from observed orbits under invariance)**.** *Suppose G is finite with $|G| = m$, the family $\{p_\theta\}$ is G-invariant, and from each independent draw X_i ($i = 1, \ldots, n$) we additionally form its orbit $\{X_i, \varphi(X_i), \ldots, \varphi^{m-1}(X_i)\}$. Because $p_\theta(\varphi^k X_i) = p_\theta(X_i)$, the orbit images are deterministic functions of X_i and carry* no additional information*: the orbital log-likelihood of the full orbit equals $m \sum_i \log p_\theta(X_i)$, whose maximizer is exactly $\hat{\theta}_n$. Consequently the effective sample size remains n (the number of independent orbits), and*

$$\sqrt{n}(\hat{\theta}_n^G - \theta_0) \xrightarrow{d} N(0, I^G(\theta_0)^{-1}), \tag{42}$$

with no $\sqrt{m}$ acceleration.

Remark 7.7 (What orbit structure does and does not buy). *The $\sqrt{nm}$ rate of a naive "treat the nm orbit points as i.i.d." analysis is an artifact: those points are not independent, and treating them as such underestimates the variance by a factor m. A genuine $\sqrt{nm}$ rate requires nm* independent *observations, which is simply more data rather than a symmetry dividend. The real benefits of the orbital approach are (a) dimension reduction on E/G and (b) variance reduction of orbit*-averaged *estimators of non-invariant functionals (Theorem 6.8), not an inflation of the sample size for a G-invariant model, for which the orbit is already a sufficient statistic ($I^G = I$, Proposition 7.3).*

8 Concentration Inequalities for Orbital Statistics

8.1 Orbital Hoeffding Inequality

Theorem 8.1 (Orbital Hoeffding). *Let $f \in C^G_{\mathrm{twin}}(E)$ be G-invariant with $f(E) \subset [a, b]$. Then:*

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}[f(X_1)]\right| \geq t\right) \leq 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right). \tag{43}$$

Proof. This is the standard Hoeffding inequality (Hoeffding, 1963), since $f(X_i)$ are i.i.d. bounded random variables. □

8.2 Orbital McDiarmid Inequality

For statistics that depend on orbits in a bounded-difference manner:

Theorem 8.2 (Orbital McDiarmid). *Let $F : (E/G)^n \to \mathbb{R}$ satisfy the bounded difference condition:*

$$|F(o_1, \ldots, o_i, \ldots, o_n) - F(o_1, \ldots, o_i', \ldots, o_n)| \leq c_i \tag{44}$$

for all $o_j, o_i' \in E/G$. Then for the orbital statistic $T_n = F(O(X_1), \ldots, O(X_n))$:

$$P(|T_n - \mathbb{E}[T_n]| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^{n} c_i^2}\right). \tag{45}$$

8.3 Concentration for Orbit-Averaged Statistics

Theorem 8.3 (Orbit-Averaged Concentration). *Let G be finite with $|G| = m$, and let $f : E \to [a, b]$ (not necessarily G-invariant). Define the orbit-averaged statistic:*

$$\bar{f}^G(X) := \frac{1}{m}\sum_{k=0}^{m-1} f(\varphi^k(X)). \tag{46}$$

Then:

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} \bar{f}^G(X_i) - \mathbb{E}[\bar{f}^G(X_1)]\right| \geq t\right) \leq 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right). \tag{47}$$

Moreover, if f varies across orbits (i.e., is not constant on orbits), then $\bar{f}^G$ typically has smaller variance than f.

9 Spectral Methods for Orbital Estimation

9.1 Twin Laplacian and Spectral Decomposition

Definition 9.1 (Twin Laplacian). *A twin Laplacian* Δ_{twin} *is a self-adjoint operator on* $L^2_{\text{twin}}(E)$ *that is* G*-equivariant:*

$$\Delta_{\text{twin}}(f \circ \varphi^n) = (\Delta_{\text{twin}} f) \circ \varphi^n. \tag{48}$$

Lemma 9.2 (Spectral Decomposition). Δ_{twin} *admits a spectral resolution:*

$$\Delta_{\text{twin}} = \int_\sigma \lambda \, dE(\lambda) \tag{49}$$

with eigenfunctions that are either G*-invariant or transform according to characters of* G*.*

9.2 Spectral Density Estimation

Definition 9.3 (Spectral Orbital Estimator). *Given the spectral decomposition of* Δ_{twin} *with eigenfunctions* $\{e_k\}$ *and eigenvalues* $\{\lambda_k\}$*, define:*

$$\hat{f}_n^{\text{spec}}(x) := \sum_{k:|\lambda_k| \le R} \hat{c}_k e_k(x) \tag{50}$$

where $\hat{c}_k = \frac{1}{n}\sum_{i=1}^n e_k(X_i)$ *and* R *is a truncation parameter.*

Theorem 9.4 (Spectral Convergence). *Under regularity conditions on the density* f *and the spectrum of* Δ_{twin}*:*

$$\mathbb{E}\|\hat{f}_n^{\text{spec}} - f\|_{L^2}^2 = O(R^{-2s}) + O\left(\frac{R^d}{n}\right) \tag{51}$$

where s *is the smoothness of* f *and* d *is the effective dimension. Optimal choice* $R \sim n^{1/(2s+d)}$ *yields rate* $n^{-2s/(2s+d)}$*.*

Remark 9.5 (Exact rate, no logarithmic factor). *The spectral estimator* (50) *is a* projection *estimator: it fits the* $O(R^d)$ *coefficients* $\hat{c}_k$ *directly, so its variance is the metric complexity* R^d/n *of the model and the balance gives the* exact *minimax rate* $n^{-2s/(2s+d)}$*, with no logarithmic factor. This is in deliberate contrast with a description-length (MDL) estimator over a quantized sieve, whose per-coefficient coding cost* $\log(1/\delta) \asymp \log n$ *inflates the rate to* $(\log n/n)^{2s/(2s+d)}$*; the MDL estimator is then near-minimax while the projection estimator above is exactly minimax. A penalized truncation choosing* R *by* $\text{pen}(R) \asymp R^d/n$ *retains the exact rate* adaptively *in* s*.*

10 Applications

10.1 Directional Statistics: The Circle

Let $E = S^1$ (the unit circle) with $G = \mathbb{Z}_m$ acting by rotations through $2\pi k/m$ (Mardia and Jupp, 2000).

Example 10.1 (Circular Data with m-fold Symmetry). *For data with* m*-fold rotational symmetry (e.g., crystal orientations), the quotient space is* $S^1/\mathbb{Z}_m \cong S^1$ *(but with* m *times shorter circumference).*

The orbital metric is:

$$d_G(O(\theta_1), O(\theta_2)) = \min_{k=0}^{m-1} |\theta_1 - \theta_2 - 2\pi k/m| \mod 2\pi. \tag{52}$$

The von Mises distribution with m-fold symmetry:

$$p(\theta; \mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos m(\theta-\mu)} \tag{53}$$

is naturally a distribution on $S^1/\mathbb{Z}_m$.

10.2 Compositional Data: The Simplex

Let $E = \Delta^{d-1} = \{x \in \mathbb{R}^d_{>0} : \sum_i x_i = 1\}$ with $G = S_d$ (permutation group) or a subgroup (Aitchison, 1986).

Example 10.2 (Exchangeable Compositions)**.** *When compositions are exchangeable (invariant under all permutations), the effective sample space is the set of order statistics, reducing dimension from $d-1$ to the number of distinct values.*

The orbital Fisher information incorporates this reduction:

$$I^{S_d}(\theta) = \frac{1}{d!} \sum_{\sigma \in S_d} I(\sigma \cdot \theta) \tag{54}$$

where the sum averages over permutations.

10.3 Physical Symmetries: Gauge Invariance

In physics, measurements are often invariant under gauge transformations (Weinberg, 1995).

Example 10.3 (Phase Estimation)**.** *Consider estimating a quantum state $|\psi\rangle$ that is only defined up to global phase: $|\psi\rangle \sim e^{i\theta}|\psi\rangle$.*

The orbit space is the projective Hilbert space $\mathbb{CP}^n$, and the natural metric is the Fubini-Study metric:

$$d_{FS}([\psi], [\phi]) = \arccos |\langle \psi | \phi \rangle|. \tag{55}$$

Estimation on $\mathbb{CP}^n$ uses the orbital framework with $G = U(1)$.

10.4 Time Series: Stationarity

For stationary time series, the distribution is invariant under time shifts (Brockwell and Davis, 1991).

Example 10.4 (Shift-Invariant Estimation)**.** *Let $X_1, \ldots, X_n$ be a stationary sequence with $G = \mathbb{Z}$ acting by shifts. The orbit-averaged estimator of the autocovariance:*

$$\hat{\gamma}(h) = \frac{1}{n-h} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X}) \tag{56}$$

exploits the shift symmetry to reduce variance compared to a single-lag estimator.

11 Computational Aspects

11.1 Computing Orbital Distances

For finite G:

$$d_G(O(x), O(y)) = \min_{g \in G} d(x, g \cdot y) = \min_{k=0}^{m-1} d(x, \varphi^k(y)). \tag{57}$$

Algorithm (Orbital Distance):

1. Input: Points $x, y \in E$, group generators.
2. Initialize $d_{\min} = d(x, y)$
3. For $k = 1, \ldots, m-1$:
   (a) Compute $y_k = \varphi^k(y)$
   (b) Update $d_{\min} = \min(d_{\min}, d(x, y_k))$
4. Return $d_{\min}$

Complexity: $O(m)$ distance computations.

11.2 Efficient Orbit-Averaged KDE

Algorithm (Orbit-Averaged KDE):

1. Input: Data $X_1, \ldots, X_n$, evaluation point x, bandwidth h.
2. For each $i = 1, \ldots, n$:
   (a) Compute orbital distance $d_i = d_G(O(x), O(X_i))$
   (b) Add contribution $K(d_i/h)$ (automatically averages over orbits)
3. Return $\frac{1}{nh^d} \sum_{i=1}^{n} K(d_i/h)$

12 Discussion

12.1 Summary of Results

We have developed a comprehensive theory of empirical estimation on orbital spaces:

1. **Orbital Glivenko-Cantelli:** Uniform convergence of orbital empirical measures with explicit rates depending on the metric entropy of the quotient space.
2. **Orbital Donsker:** Functional central limit theorem for orbital empirical processes.
3. **Equivariant KDE:** Kernel density estimation using equivariant kernels achieves optimal rates on quotient spaces, with orbit-averaging providing variance reduction.
4. **Orbital MLE:** Maximum likelihood on quotient spaces with asymptotic normality governed by orbital Fisher information.
5. **Concentration:** Hoeffding and McDiarmid inequalities for orbital statistics.

12.2 Benefits of the Orbital Approach

1. **Dimension reduction:** Working on E/G reduces the effective dimension when G is large.
2. **Variance reduction:** Orbit-averaging reduces variance by a factor up to $|G|$.
3. **Equivariance:** Estimators automatically respect the symmetry structure.
4. **Interpretability:** Results are stated in terms of orbits, which are often the natural objects of interest.

12.3 Open Problems

1. Optimal estimation of the group G itself from symmetric data
2. Minimax rates for density estimation on general quotient spaces. *(Now largely established: the exact rate $n^{-2s/(2s+d_{\text{eff}})}$ holds on a d_{eff}-dimensional quotient—Fano lower bound matched by a projection estimator, exactly and adaptively. What remains open is the fully stratified orbifold quotient and the heavy-tailed multiplicative regime.)*
3. Bootstrap methods adapted to orbital structure
4. High-dimensional orbital estimation with $\dim(E/G) \to \infty$
5. Non-parametric hypothesis testing on quotient spaces

13 Conclusion

This article has established the foundations of empirical estimation on orbital spaces, where the sample space is partitioned by the action of a discrete group. The key insight is that statistical procedures should respect the orbital structure, leading to equivariant estimators that exploit symmetry for improved performance.

The orbital framework provides:

- A natural setting for statistics with symmetries
- Dimension reduction through quotient spaces
- Variance reduction through orbit-averaging
- Connection to the Twin Space approximation theory

A Covering Numbers for Orbital Function Classes

Lemma A.1 (Orbital Covering Number). *Let $\mathcal{F}^G = \{f \in C^G_{\text{twin}}(E) : \|f\|_{\text{Lip},G} \le L\}$ be the class of G-invariant L-Lipschitz functions. If $(E/G, d_G)$ has covering dimension d, then:*

$$\log N(\epsilon, \mathcal{F}^G, \|\cdot\|_\infty) \le C \cdot \left(\frac{L}{\epsilon}\right)^d \tag{58}$$

for some constant C depending on E/G.

B Explicit Formulas for Common Groups

B.1 Cyclic Group $\mathbb{Z}_m$

For $E = \mathbb{R}^2$ and $G = \mathbb{Z}_m$ acting by rotation through $2\pi/m$:

$$\varphi^k(x, y) = (x\cos(2\pi k/m) - y\sin(2\pi k/m), x\sin(2\pi k/m) + y\cos(2\pi k/m)). \tag{59}$$

The orbital distance:

$$d_G(O(z_1), O(z_2)) = \min_{k=0}^{m-1} |z_1 - e^{2\pi ik/m} z_2| \tag{60}$$

in complex notation.

B.2 Symmetric Group S_n

For $E = \mathbb{R}^n$ and $G = S_n$ acting by permutation:

$$\sigma \cdot (x_1, \ldots, x_n) = (x_{\sigma^{-1}(1)}, \ldots, x_{\sigma^{-1}(n)}). \tag{61}$$

The orbit is the set of all permutations of $(x_1, \ldots, x_n)$. The orbital distance uses sorted vectors:

$$d_G(O(x), O(y)) = \|x_{(\cdot)} - y_{(\cdot)}\| \tag{62}$$

where $x_{(\cdot)}$ denotes the sorted version of x.

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